

# Real-World Architecture Example: Modernizing a Retail Commerce Platform on AWS

## A Serverless, Cloud-Native Approach

A practical example of the value of cloud computing, and serverless architecture in particular, can be seen in a mid-sized retail organization modernizing its e-commerce platform. The company needs to support highly variable customer demand, maintain high availability during seasonal traffic spikes, improve the timeliness of business insights, and start using artificial intelligence for personalization. These needs expose the limits of traditional on-premises infrastructure, including rigid capacity planning, expensive disaster recovery, and limited flexibility for innovation.

### Business Challenge

The organization's existing on-premises environment is designed around peak demand, which means resources are underused during normal operations and require significant capital investment. Scaling for high-traffic events requires advance procurement and long lead times, creating both financial and operational risk. Reporting is also delayed because data systems are fragmented, and the organization does not have an efficient way to experiment with AI-driven capabilities such as product recommendations.

### AWS Serverless Architecture Approach

The modernized architecture is designed as a serverless, cloud-native platform on AWS. This removes the need to manage provisioned infrastructure directly. Each layer of the architecture uses AWS managed services in serverless configurations, so costs align with actual usage and scaling happens automatically as demand changes.

User traffic is routed through Amazon CloudFront and AWS WAF for global content delivery and protection. Application workloads run on containerized services using Amazon ECS with AWS Fargate, the serverless compute engine for containers. This allows the application to scale automatically based on demand without requiring EC2 instance management.

Transactional data is managed with Amazon Aurora Serverless v2, which automatically scales relational database capacity based on application demand and bills only for the capacity consumed. Amazon ElastiCache Serverless improves application responsiveness for session management and frequently accessed data, scaling quickly without requiring cluster configuration or node type selection.

Event-driven components use Amazon SQS, Amazon EventBridge, and AWS Lambda to decouple services and improve resilience. These services are fully serverless and scale transparently with message and event volume.

Operational and analytical data is centralized in Amazon S3. AWS Glue provides serverless ETL, and Amazon Redshift Serverless provides on-demand data warehouse capacity for business intelligence reporting. This allows analytics workloads to scale automatically while charging only for compute

used during query execution.

The architecture also supports AI workloads through Amazon Bedrock, which provides fully managed, serverless access to foundation models. This makes it easier to experiment with and deploy intelligent features such as product recommendations and personalization.

Security, monitoring, and governance are managed through AWS Identity and Access Management (IAM), AWS Key Management Service (KMS), Amazon CloudWatch, and AWS CloudTrail. These services help provide visibility, compliance, and operational control across the platform.

## Business Outcomes

**Cost Optimization:** The architecture eliminates the need for over-provisioned infrastructure by aligning resource consumption, including database, cache, and analytics compute, with actual demand. Serverless billing also reduces idle cost compared with elastic but still provisioned environments.

**Scalability and Resilience:** Every tier scales automatically, from containers to databases to analytics, so the platform can handle peak traffic while maintaining high availability through managed, multi-AZ services.

**Operational Efficiency:** The platform removes most undifferentiated infrastructure management. Engineering teams can focus on delivering business value instead of patching nodes, sizing clusters, or managing capacity.

**Speed and Agility:** Automation and Infrastructure as Code tools such as AWS CloudFormation and the AWS Cloud Development Kit (CDK) make it faster to develop, deploy, and recreate environments in a repeatable, version-controlled way.

**Data-Driven Decision Making:** Amazon Redshift Serverless and AWS Glue enable near real-time analytics and reporting, helping the business respond faster without dedicated data warehouse infrastructure.

**Innovation Enablement:** On-demand access to foundation models through Amazon Bedrock, along with specialized serverless compute, lowers the barrier to adopting AI and advanced analytics. Teams can experiment quickly without upfront infrastructure investment.

## Strategic Alignment

By adopting a serverless, cloud-native architecture on AWS, the organization turns infrastructure from a fixed operational constraint into a strategic enabler. Each component in the platform scales on demand, bills only for what it consumes, and requires no server-level management, freeing engineering capacity for differentiated product and customer experience work.

The platform supports global scale, improves the customer experience through AI-powered personalization with Amazon Bedrock, and positions the business to respond quickly to market opportunities. Some specialized workloads may still remain on-premises or in hybrid configurations, but a serverless-first, cloud-native approach provides a more flexible, cost-effective, and

innovation-friendly foundation for most workloads.